ARel CAMPUS INSTITUTE OF TECHNOLOGY

SCHOOL ENROLMENT, ATTENDANCE AND STUDENT RETENTION ANALYSIS

SQL DATA ANALYSIS PROJECT REPORT

Project detail	Information
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Course	Data Analysis
Database / Tool	school_data / MySQL Workbench
Dataset	2026 practice dataset
Date	6 November 2026

1. Executive Summary

This project used SQL to analyse student enrolment, attendance, student movements and data quality. The dataset recorded 119 Term 1 students. Attendance varied across classes and months, while four transfers and two withdrawals were recorded.

2. Business Problem

School administrators need clear information to monitor enrolment, identify low attendance and maintain accurate student records. This analysis uses SQL to highlight areas that may need attention.

3. Key Findings

1. Enrolment by gender

SQL counted distinct Term 1 students and grouped them by gender.

Gender	Students	Share
Girls	63	52.94%
Boys	56	47.06%
Total	119	100%

Finding: There were seven more girls than boys.

2. Enrolment by class

SQL grouped students by class.

Class	Students
Grade 7 East	13
Grade 7 West	19
Grade 8 East	23
Grade 8 West	22
Grade 9 East	21
Grade 9 West	21

Finding: Grade 8 East had the most students (23); Grade 7 East had the fewest (13).

3. Overall attendance

SQL divided Present records by all attendance records.

Finding: 3,555 of 4,660 records were Present, giving an attendance rate of 76.29%.

4. Attendance by class

SQL calculated and ranked the Present rate for each class.

Class	Rate
Grade 8 West	68.64%
Grade 9 West	72.07%
Grade 9 East	78.78%
Grade 8 East	79.13%
Grade 7 West	79.29%
Grade 7 East	82.88%

Finding: Grade 8 West had the lowest rate; Grade 7 East had the highest.

5. Monthly attendance

SQL grouped attendance by month.

Month	Rate
January	76.47%
February	77.55%
March	77.67%
April	72.05%
May	75.44%
June	74.56%
July	77.37%
August	77.19%

Finding: March was highest (77.67%) and April lowest (72.05%).

6. Transfers and withdrawals

SQL counted student movement records.

Finding: The dataset recorded 4 transfers and 2 withdrawals.

7. Attendance by term

SQL calculated attendance separately for each term.

Finding: Term 1 was 76.81% and Term 2 was 75.75%, a decrease of 1.06 percentage points.

8. Data quality

SQL checked missing details, duplicate records and unmatched student or class records. The selected checks returned no problems in the fields examined.

4. Recommendations

- Investigate the low attendance in Grade 8 West.
- Review the attendance drop between March and April.
- Monitor attendance regularly by class and term.
- Keep enrolment, transfer and withdrawal records up to date.
- Continue checking for missing and duplicate records.

5. Conclusion

SQL helped summarise enrolment, attendance, student movements and data quality. The findings show where further monitoring may be useful. Since this is a practice dataset, the results should be verified before being used for real school decisions.

Note: The overall attendance rate (76.29%) and term rates use different query scopes; they should not be treated as directly comparable.